Detecting Heart Abnormality Using ECG with CART

Machine Learning Mini-Project

Team 31:

Nihira Hassan - PES2UG24AM106

Shreya Raghuraj- PES2UG24AM154

1. Problem Statement

Electrocardiogram (ECG) records capture electrical activity of the heart and can contain patterns associated with different arrhythmia classes. This project builds a Classification and Regression Tree (CART) classifier using the UCI Arrhythmia dataset. The primary task is multiclass prediction of the original class labels. A baseline decision tree is compared with a hyperparameter-tuned tree using the same held-out test set.

2. Dataset and Preprocessing

The supplied dataset contains 452 records and 280 columns: 279 input attributes and one target label. Attribute 14 contains 376 missing values (83.2% of records), so it is removed, leaving 278 input features. The observed target labels are 1–10, 14, 15, and 16. Class 1 denotes the normal class in the original dataset; the notebook retains the original multiclass labels for the primary experiment. Remaining numerical missing values are handled by median imputation within a scikit-learn Pipeline.

3. Methodology and Models

The data is split into 70% training and 30% testing using stratified sampling with random_state=42 (316 training records and 136 test records). The baseline model is a DecisionTreeClassifier using Gini impurity and default complexity settings. The comparison model uses GridSearchCV with 3-fold StratifiedKFold and macro-F1 as the selection metric. The search explores maximum depth, minimum samples per leaf, class weighting, and cost-complexity pruning (ccp_alpha). The best tuned settings were max_depth=12, min_samples_leaf=2, class_weight='balanced', and ccp_alpha=0.01.

4. Results

Metric	Baseline multiclass CART	Tuned multiclass CART
Accuracy	69.12%	55.88%
Balanced accuracy	36.98%	39.96%
Macro F1-score	0.3657	0.3666

The baseline model achieved higher overall accuracy and is retained as the primary model. The tuned model achieved slightly higher balanced accuracy and macro F1, indicating a small improvement in performance averaged across classes, but its overall accuracy was lower. Rare classes have very few examples, which limits the reliability of class-wise and cross-validation estimates.

5. Implementation and Demonstration

The notebook uses Python, pandas, NumPy, Matplotlib/Seaborn, and scikit-learn. It evaluates both models, plots the baseline confusion matrix and decision-tree visualization, and visualizes feature importance. In the recorded sample demonstration, row index 222 has actual class 6; both baseline and tuned CART predict class 6. This is a demonstration on a dataset record, not an independent clinical prediction.

6. Conclusion and Limitations

This project demonstrates an end-to-end CART workflow for multiclass arrhythmia classification, including missing-value handling, stratified evaluation, hyperparameter search, and model interpretation. The dataset is small and imbalanced, and several classes have very few examples, so results may vary with the split. Future work could evaluate repeated stratified splits, obtain more representative data, and compare additional models while preserving leakage-safe preprocessing. This is an educational prototype only and must not be used for medical diagnosis or treatment decisions.

References

https://cs229.stanford.edu/proj2014/Ben Zhou, Gaspar Garcia, Paurakh Rajbhandary, Detecting Heart Abnormality using ECG with CART.pdf

UCI Machine Learning Repository, Arrhythmia dataset: https://archive.ics.uci.edu/dataset/5/arrhythmia